

SAFETY DATA SHEET



Date issued : 05/06/2020
 SDS number : SCC-P78
 Date revised : 03/04/2025
 Revision number : 4

P78 Electric Resistor Cement, Powder

1. Identification

Product code: P78
Product identifier: P78 Electric Resistor Cement, Powder
Product description: Electric Resistor Cement, Powder
Product formulation name: P78 Electric Resistor Cement, Powder
Chemical family: Sodium Silicate Cement

Manufacturer / Supplier

Sauereisen
 160 Gamma Drive
 Pittsburgh, PA 15238
Emergency contact: John Kozak
Emergency Phone: (800)444-8235
Alternate Contact: Rob Bowman
Customer Service: 412 963-0303
E-Mail: jakozak@sauereisen.com

Emergency telephone number (24 hour)

Poison Control Center (Medical): 1-800-222-1222
 CHEMTREC (US Transportation): 1-800-424-9300
 CHEMTREC (Canada Transportation): 1-703-527-3887

2. Hazard identification

Classification of the substance or mixture**Health hazards:**

Acute Toxicity (Inhalation), Category 4
 Eye Irritation, Category 2B
 Skin Irritation, Category 2
 Target Organ Toxicity (Repeated exposure), Category 2
 Carcinogenicity, Category 1B

Label elements

Exclamation
 mark



Health
 hazard

Signal word: DANGER

Hazard statement(s)

H315: Causes skin irritation.
 H320: Causes eye irritation.
 H333: May be harmful if inhaled.
 H350: May cause cancer .
 H373: May cause damage to lungs or kidneys through prolonged or repeated exposure via inhalation.

Precautionary statement(s)**Prevention:**

P201: Obtain special instructions before use.
 P202: Do not handle until all safety precautions have been read and understood.
 P260: Do not breathe dust/fume/gas/mist/vapours/spray.
 P264: Wash ... thoroughly after handling.
 P270: Do not eat, drink or smoke when using this product.
 P271: Use only outdoors or in a well-ventilated area.

P280: Wear protective gloves/protective clothing/eye protection/face protection.

Response:

P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P302+P352: IF ON SKIN: Wash with plenty of water/...

P314: Get medical advice/attention if you feel unwell.

P332+P313: If skin irritation occurs: Get medical advice/attention.

P322: Specific measures (see ... on this label).

P362: Take off contaminated clothing.

P304+P340: IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P312: Call a POISON CENTER/doctor/...if you feel unwell.

Storage:

P405: Store locked up.

Disposal:

P501: Dispose of contents/container in accordance with local/national regulations.

Potential health effects

Eye: Contact may cause eye irritation.

Skin: Itching or burning will occur with repeated or prolonged contact. May exacerbate existing skin conditions.

Inhalation: Dust or vapor irritating to eyes and respiratory tract.

3. Composition/information on ingredients

Chemical name	% w/w	CAS No.
Silica, Crystalline	< 50	14808-60-7
Aluminosilicate	< 50	1327-36-2
Aluminum Oxide	< 10	1344-28-1

4. First-aid measures

Eye: Immediately flush eyes with plenty of water. Get medical attention, if irritation persists.

Skin: Wash contact area thoroughly with soap and water. Remove contaminated clothing. Launder before reuse. Seek medical attention if erythema develops.

Ingestion: If swallowed, do not induce vomiting. Give large quantities of water. Seek medical attention immediately. Never give anything by mouth to an unconscious person. Advise physician of possible fluoride exposure.

Inhalation: If difficulty breathing, move to fresh air once. Apply artificial respiration if breathing has stopped. Seek medical attention.

Most important symptoms and effects, both acute and delayed

Eye: Causes pain, redness and tearing.

Skin: Contact causes skin irritation.

Inhalation: Headache, nausea, and irritation to nose and throat. Prolonged or repeated exposure may cause asthma.

Chronic effects: The adverse health effects-- silicosis, lung cancer, autoimmune and chronic kidney diseases, tuberculosis and non-malignant respiratory diseases-- are chronic effects.

Prolonged exposure to fluoride over years may produce an embrittlement and densification of bones and increase calcification of ligaments.

Indication of immediate medical attention and special treatment needed, if necessary: Further treatment may be necessary. Contact local poison control center. Rinse mouth.

5. Fire-fighting measures

Flammable class: NA = Not Applicable

Flame propagation or burning rate of solids: None

General hazard: NA = Not Applicable

Suitable extinguishing media: NA = Not Applicable

Hazardous combustion products: NA = Not Applicable

Explosion hazards: Not sensitive to mechanical impact or static discharge

Fire fighting procedures: NA = Not Applicable

Fire fighting equipment: NA = Not Applicable

Fire explosion: NA = Not Applicable

Sensitivity to static discharge: None

Sensitivity to mechanical impact: None

Hazardous decomposition products: NA = Not Applicable

6. Accidental release measures

Small spill: Clean-Up Procedure: Spill area may become slippery; use care to avoid falls. Flush spill area with plenty of water.

Large spill: For spill greater than 100 gallons; isolate, dike and store discharged material, if possible. Collect with dry sand, clay or other absorbent. Flushed cleaned areas with water. Observe Environmental regulations. Wear PPE - gloves, rubber boots, and safety glasses.

Special protective equipment: Wear gloves, rubber boots, and safety glasses.

7. Handling and storage

General procedures: Avoid contact with eyes, skin, and clothing.

For industrial use only!

Do not take internally.

May cause irritation.

Wash thoroughly after handling.

Precautions for safe handling: Wear protective safety glasses, gloves and protective clothing.

Conditions for safe storage: Store in a cool, dry place.

Keep container closed when not in use.

Keep away from food and drinking water.

Always mix well before using.

8. Exposure controls/personal protection

Exposure controls

Control parameters				
Occupational exposure limit values				
Chemical name	Type		ppm	mg/m ³
Silica, Crystalline	OSHA PEL	TWA		5
		STEL	0.05	
	ACGIH TLV	TWA		0.025
	Supplier OEL	TWA	NL	NL
		STEL	NL	NL
Aluminosilicate	OSHA PEL	TWA		10
	ACGIH TLV	TWA		2
Aluminum Oxide	ACGIH TLV	STEL	10	

Individual protection measures, such as personal protective equipment

Eye / face protection: Safety glasses with side shields, chemical resistant goggles, or face shield. Contact lenses should not be worn.

Skin protection - hand protection: Permeation resistant gloves (butyl rubber, nitrile rubber or poly vinyl alcohol). Please note that

PVA degrades in water. Cover exposed skin as possible with appropriate clothing. If skin creams are used, keep area covered by cream to a minimum.

Respiratory protection: This product contains encapsulated silica. By OSHA letter of interpretation, the silica is not considered respirable in either the cement paste form, or cured cement form. However, if the cured cement is polished, ground, or chipped during processing, handling, or use, the silica may be released as an airborne respirable particle. In these Instances appropriate personal protective equipment and local ventilation controls must be employed.

9. Physical and chemical properties

Appearance: Powder

Color: White

Odor: No appreciable odor.

Odor threshold: Not Available

pH: 11.5

Flash point: NA = Not Applicable

Lower explosion limit / flammability limit: 0

Upper explosion limit / flammability limit: 0

Vapor pressure: Vapor is water

Relative vapor density: Same as water vapor.

Relative density: 1

Auto-ignition temperature: NA = Not Applicable

Percent volatiles: NA = Not Applicable

10. Stability and reactivity

Reactivity: Yes

Dangerous polymerization: No

Chemical stability: Stable under normal conditions of use and storage.

Hazardous decomposition products: Silica will dissolve in hydrofluoric acid and produce a corrosive gas - Silicon tetrafluoride.

Incompatible materials: Avoid contact with acids, and acid vapors.

11. Toxicological information

Acute toxicity

Acute dermal toxicity LD₅₀: No data available.

Acute oral toxicity LD₅₀: No data available.

Acute inhalation toxicity LC₅₀: No data available.

Serious eye damage / irritation: Eye, Skin and Inhalation Irritant.

Germ cell mutagenicity: Not Available

Carcinogenicity

IARC: Silica is listed as having sufficient evidence to be a carcinogen in humans and in experimental animals, for the carcinogenicity of quartz and cristobalite. The overall IARC evaluation was that crystalline silica inhaled in the form of quartz or cristobalite from occupational sources is carcinogenic to humans (group 1).

NTP: The National Toxicology Program, in it's Ninth Annual report on Carcinogens, classified "silica, crystalline (respirable)" as a known human carcinogen.

OSHA: Crystalline Silica (Quartz) is not regulated by the US Occupational Safety and Health Administration as a carcinogen.

Notes: Silica is listed by IARC and NTP as having sufficient evidence to be a carcinogen in humans and in experimental animals for the carcinogenicity of quartz and cristobalite. The overall IARC evaluation was that crystalline silica inhaled in the form of quartz or cristobalite from occupational sources is carcinogenic to humans (group 1).

Reproductive toxicity: Not Available

Specific Target Organ Toxicity - single exposure:

- Nephrotoxicity - Recent studies suggest that exposure to respirable crystalline silica or that the disease silicosis is associated with the increased incidence of kidney disorders.

General comments:

ADDITIONAL INFORMATION: Crystalline Silica (Quartz)

- Silicosis - The major concern is silicosis caused by the inhalation of respirable crystalline silica dust. Silicosis can exist in several forms, chronic (or ordinary), accelerated, or acute.
- Scleroderma - There is evidence that exposure to respirable crystalline silica or silicosis is associated with incidence of scleroderma of the lungs.
- Tuberculosis - Individuals with silicosis are at risk to develop tuberculosis, if exposed to persons with tuberculosis.

12. Ecological information

Aquatic toxicity, both acute and chronic: Sodium Silicate has low toxicity. TLM 96 hr in mosquitofish is 2,320 mg/L. High pH (alkalinity) of undiluted material is harmful to aquatic life.

96-hour LC₅₀: ≤ 5000

Environmental data: Crystalline silica (quartz) is not known to be an environmental hazard.

13. Disposal considerations

Disposal methods: Comply with federal, state and local regulations. If approved, flush with water to chemical sewer. For large quantities, neutralize with dilute acid and landfill solids or flush neutral liquid to sewer with plenty of water.

14. Transport information

USA Department of Transport Regulations (DOT)

UN proper shipping name: Not Regulated

Technical name: NA = Not Applicable

Transport hazard class(es): Not Regulated

ADR / RID - road / rail

UN proper shipping name: Not Regulated

ICAO / IATA - air

UN proper shipping name: Not Regulated

IMO / IMDG - International

UN proper shipping name: Not Regulated

Canadian Transport of Dangerous Goods Regulations (TDG)

UN proper shipping name: Not Regulated

15. Regulatory information

UNITED STATES

Dot label symbol and hazard classification



Toxic

R36/37/38: Irritating to eyes, respiratory system and skin.

R40: Limited evidence of a carcinogenic effect.

R48/20: Harmful : danger of serious damage to health by prolonged exposure through inhalation.

S22: Do not breathe dust.

S24/25: Avoid contact with skin and eyes.

S26: In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

S36/37/39: Wear suitable protective clothing, gloves and eye/face protection.

S38: In case of insufficient ventilation, wear suitable respiratory equipment.

SARA Section 311/312 Hazard Categories

311/312 Health hazards: Chronic. Carcinogen. Irritant.

313 reportable ingredients: There are no listed chemicals above detection limits in this compound.

EPCRA Section 313 Toxic Chemicals

Chemical name	% w/w	CAS No.
Aluminum Oxide	< 10	1344-28-1

Title III notes: None above detection limits.

CERCLA Hazardous Substances and Reportable Quantities (RQ)

CERCLA regulatory: None

TSCA (The Toxic Substances Control Act)

Chemical name	CAS No.
Silica, Crystalline	14808-60-7
Aluminosilicate	1327-36-2
Aluminum Oxide	1344-28-1

TSCA Status: Components are included in the EPA Toxic Substances Control Act (TSCA) Chemical Substances Inventory.

Regulations

State regulations:

Massachusetts Toxic Use Reduction Act- Silica, Crystalline (respirable size, <10microns) is toxic for purposes of the Massachusetts Toxic Use Reduction Act

Pennsylvania Worker and Community Right to Know Act- Quartz is a hazardous substance under the act, but it is not a special hazardous substance or an environmental hazardous substance.

California Inhalation Reference Exposure Level (REL)- California established a chronic REL of 3 ug for silica crystalline, respirable). A chronic REL is an airborne level of a substance at or below which no adverse health effects are anticipated in individuals indefinitely exposed to the substance at that level.

California Proposition 65: Known to the State of California to cause cancer or reproductive toxicity and subject to warning and discharge requirements under the "Safe Drinking Act of 1986".

It has not been determined and cannot be ascertained that this product would not expose users to the listed chemicals at the very low level prescribed in the regulations. Therefore, it is the user's responsibility to determine if the percent of the hazardous / carcinogenic ingredients listed elsewhere in the SDS comply with State of California regulations.

CANADA

WHMIS Hazard Symbol and Classification



Toxic

R36/37/38: Irritating to eyes, respiratory system and skin.

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S38: In case of insufficient ventilation, wear suitable respiratory equipment.

WHMIS Classification: Class D, Division 2, Subdivision A: Materials cause other toxic effects, very toxic material.

Domestic Substances List (DSL) / Non-Domestic Substances List (NDSL): Components included on inventory

16. Other information

Prepared by: John A Kozak **Date revised:** 03/04/2025

Revision summary: This SDS replaces the 07/22/2020 SDS. Revised: **Section 2:** Classification of the substance or mixture, Label elements, Precautionary statement(s).

HMIS rating

Health	*	1
Flammability		0
Physical hazard		0
Personal protection		E